Web Services

- SEng5071

Course Information

- Course Contents:
  - 6 chapters
- Lab practice:
  - PHP/JAVA, MySQL, JavaScript, HTML5, XML or JSON
  - IDE : NetBeans
- Assessment
  - Group lab assignment
  - Group course project
    - Group members list— (October 13,2026)
    - Title or area will be given using lottery
    - Requirement, data base, class, object and method designs will be from the instructor
    - Final report submission and presentation –

CHAPTER 1:

Web Services Overview

Outline:

- Introduction of Web services
  - What are web services?
  - What is a service-oriented architecture (SOA)?
  - Services
  - Connections
- Examples of Web Services
- Why Use Web Services?
- Web Service Considerations
- Service-Oriented Architecture (SOA) Overview
- Relationship Of Web Services And SOA
- Identification And Design Of Services

Introduction of Web services

- What are web services?
  - So far, we've been designing a website to be consumed by browsers
  - Now we turn our attention to making data and functionality available to other programs
  - Increasingly, the Internet is no longer a collection of isolated websites,
    - but a true web: websites communicate freely with each other in order to provide a richer experience for the user
  - It's a programmer's dream come true: the Internet is becoming as accessible to your code as it has traditionally been to real people

Introduction of Web services (CONTD…)

- What are web services? (CONTD…)
  - One of the inevitable truisms of enterprise applications is that they are not islands
  - You may be focused on solving a particular business problem, but in order to do that you won't be able to capture all the data you need yourself, or develop all the processing
  - Even if you had the time, that data and processing is being done elsewhere, and duplication is both wasteful and leads to messy inconsistencies
  - As a result, almost all enterprise applications need to communicate with other applications
  - These foreign systems are often not in the same organization, but are provided by some third-party organization

Introduction of Web services (CONTD…)

- What are web services? (CONTD…)
  - For many years, one of the hardest parts of this kind of collaboration was just to get some kind of communication path
  - Often these applications were written on different platforms, with different languages, on different operating systems supporting different communication protocols
  - But in the past decade, the web has appeared as a solution to the connection problem
  - Almost all systems can open port 80 and talk text over it

Introduction of Web services (CONTD…)

- What are web services? (CONTD…)
  - But that still leaves many questions around how they should talk
  - Should they use an RPC-style API, a message-oriented API, or this fashionable REST stuff?
  - Should logic be embedded in services directly or delegated to underlying objects?
  - How can we change services that are already in use without breaking clients?
  - You can add a web service to your app
  - The term “web service” is a general term that means any application programming interface (API) that’s accessible over HTTP

Introduction of Web services (CONTD…)

- What are web services? (CONTD…)
  - Web service is defined as a means to connect services together
  - A service is software that performs some computing function and has some type of underlying computer system
  - According to W3C,
    - A Web service is a software system designed to support interoperable machine-to-machine interaction over a network
    - It has an interface described in a machine-processable format (specifically WSDL)
    - Other systems interact with the Web service in a manner prescribed by its description using SOAP messages, typically conveyed using HTTP with an XML serialization

Introduction of Web services (CONTD…)

- What is a service-oriented architecture (SOA)?
  - SOA is a design and development style and, the result of that development is commonly referred to as an architecture
  - Service-oriented architecture is a way to design, implement, and assemble services to support or automate business functions
  - Various Web services can be used to connect services
  - The assembly of services—internal and external to an organization—makes up a service-oriented architecture (SOA)

Introduction of Web services (CONTD…)

- What is a service-oriented architecture (SOA)? (CONTD…)
  - Figure 3.2 illustrates a basic service-oriented architecture
  - It shows a service consumer at the right sending a service request message to a service provider at the left
  - The service provider returns a response message to the service consumer
  - The request and response connections are defined in some way that is understandable to both the service consumer and the service provider

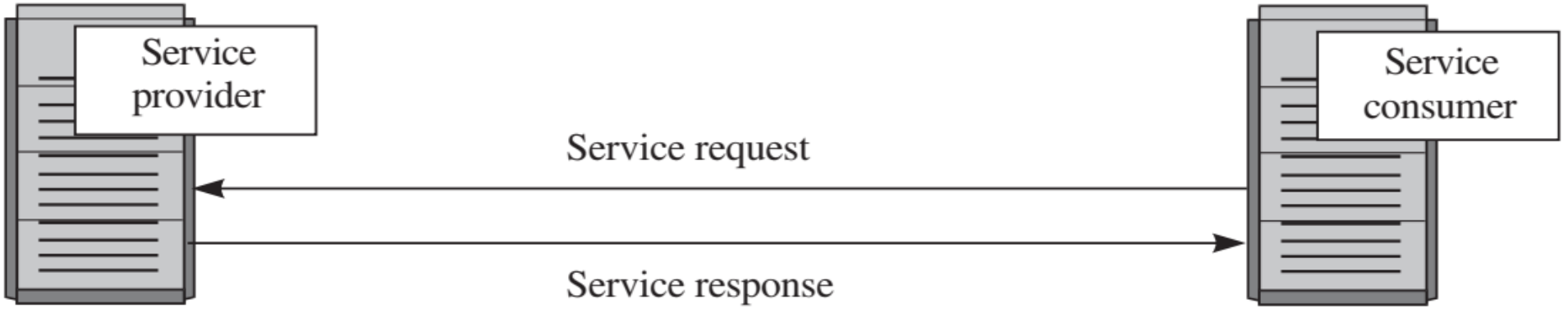


Figure 3.2 SOA basics.

Introduction of Web services (CONTD…)

- Services
  - A service is software and hardware
  - One or more services support or automate a business function
  - The intent is that a service can be used in multiple ways (often referred to as reusability)
  - There are two types of services: atomic and composite
    - An atomic service is a well-defined, self-contained function that does not depend on the context or state of other services
    - A composite service is an assembly of atomic or other composite services
      - A service within a composite service may depend on the context or state of another service that is also within the same composite service

Introduction of Web services (CONTD…)

- Services (CONTD…)
  - standard services as commodities
    - Organizations will eventually evolve standard capabilities of CRM, enterprise resource planning (ERP), and other services
    - These standard services could be seen as commodities
    - We may see these services come in various forms
    - What does this mean for software development?
    - It means fewer people writing software and more organizations buying software or renting access to software rather than building it

Introduction of Web services (CONTD…)

- Connections
  - Web services provide the means of connecting services
  - Connections such as Web services are part of the inevitable evolution of interconnectedness
  - Consider how we can now exchange email among disparate products
  - we could not do that at one time, we now take it for granted
  - This e-mail exchange is possible because of standards
  - Connections like Web services can also be taken for granted because of sets of standards that were developed

Examples of web services

- Using object-oriented programming in PHP as an example
  - say you build an application using objects
  - The classes for the objects were well thought out, so each performs operations for specific areas of functionality

  - Another area of the company is working on a separate application and
  - ends up needing to access functionality from the first application

  - On top of that, this new application isn't even written using PHP so cannot reuse any code natively

Examples of web services (CONTD…)

- Using object-oriented programming in PHP as an example (CONTD…)
  - The bruteforce method would be to have this new application duplicate the logic the PHP application does
  - This, however, presents problems if the logic were to change in the PHP application
  - The other application would need to also change its logic or
  - face the problem that it no longer works correctly,
  - which could lead to a variety of problems within the company, including data corruption

Examples of web services (CONTD…)

- Using object-oriented programming in PHP as an example (CONTD…)
  - With SOA using Web services , the PHP application can expose the functionality of its classes via a service
  - The other application can access the functionality of the PHP application
  - Through a common protocol and descriptive messaging
  - For example,
    - a daemon, which is a process waiting for invocation to perform a task,
    - is written in PHP and run via the PHP command-line interpreter (CLI)

Examples of web services (CONTD…)

- Using object-oriented programming in PHP as an example (CONTD…)
  - The daemon accepts connections via Transmission Control Protocol/Internet Protocol (TCP/IP) and
  - processes requests based on the messages it receives, which are written in some company-standardized text language
  - This text language describes the class to access, the function to call, the arguments, and their values needed by the function
  - The outside application then connects to the daemon, sends its message, and receives some response
  - Because the task was an external process, the calling application does not care how it was done, just that it was performed

Examples of web services (CONTD…)

- adding a REST API to the Meadowlark Travel website
  - Another example of web service can be by adding a REST API to the Meadowlark Travel website
  - To encourage travel to Oregon, Meadowlark Travel maintains a database of attractions, complete with interesting historical facts
  - An API allows the creation of apps that enable visitors to go on self-guided tours with their phones or tablets:
    - if the device is location-aware, the app can let them know if they are near an interesting site

Examples of web services (CONTD…)

- for planning and taking a business trip
  - This is the story of C.R., which is short for Connected Representative
  - In this story, C.R. is about to take a business trip to Europe
  - This trip will involve
    - visiting multiple customers in different cities over three or four days and
    - responding to routine tasks from the office
  - At one time, C.R. carried a cell phone and a laptop on business trips
  - Nowadays, C.R. carries just a smartphone

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - On this trip, C. R. will also wear his regular eyeglasses that are augmented with a heads-up display, an earpiece, and a camera
  - The eyeglasses communicate with his smartphone
  - To start planning his trip, C.R. uses a smartphone application that is part of his virtual personal assistant (VPA)
  - He asks the VPA to find all customers near each stop in his trip and
  - to rank them based on criteria from his organization's business intelligence (BI)/analytics system

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - There are two components to C. R.'s VPA
    - One component is on his smartphone, which has already been mentioned
    - Another component exists in the public cloud
      - This means C. R. can access his VPA using other devices (e.g., someone else's phone or a desktop) should he desire
      - It also allows the VPA to help manage his life even if C. R.'s smartphone is turned off or unreachable
      - Many of the travel arrangements described here are handled by the VPA component in the public cloud

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - The VPA takes advantage of application programming interfaces (APIs) that use standard vocabularies (the data and the names to use when describing the data)
  - Airlines, hotel chains, car rental companies, restaurant reservation systems, calendaring systems, and many other services on the Internet have agreed to use standard vocabularies
  - Recently, C. R.'s organization added similar APIs to the repository it maintains in its virtual private cloud so that employees' VPAs can interact with the repository

Examples of web services (CONTD…)

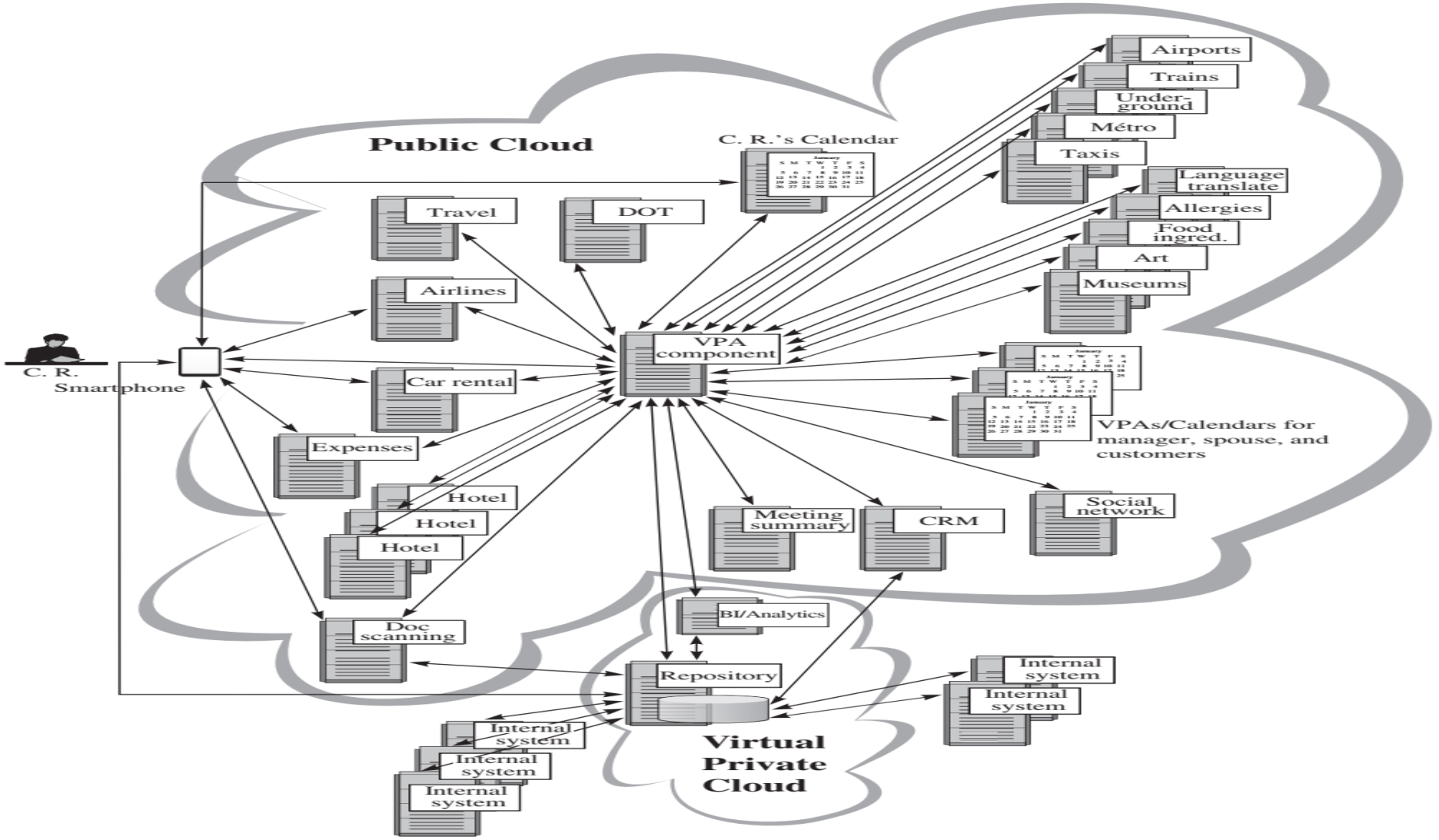
- for planning and taking a business trip (CONTD…)
  - There obviously need to be agreements and standards among organizations to make this level of data interchange possible
  - This technology and the standards make it possible for C.R. to be “connected” on his business trip
  - Many services and supporting technologies came together in the business trip story,
    - including documents and customer data from internal systems,
    - an external customer relationship management (CRM) service,
    - calendar services,
    - a travel website,
    - a car rental service, and more

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - Virtual personal assistants (VPAs) are central to this story
  - C. R's VPA is software and, in our story, it uses artificial intelligence to reason and learn from experience
  - The wealth of connections available on the Internet makes it possible to create various types of VPAs that can take advantage of those connections
  - Figure 2.1 below illustrates that C.R.'s VPA has a component that is a service much like other services in the cloud
    - C. R.'s smartphone interacts with the VPA component in the cloud
    - The VPA service in the cloud acts independently

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)



igure 2.1 Services and data interchange related to C. R.'s business trip.

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - C.R.'s VPA managed the business trip
  - It was able to gather information from different services,
    - make travel arrangements,
    - monitor data feeds,
    - "jump" in at the last moment when needed, and
    - provide C. R. with just the information he needed
  - Thereby, C. R. was able to do what he does best (talking with customers)
  - without overburdening him with having to manage his trip himself or
  - sort through a flood of information

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - One illustration of the VPA's role with various services was
    - how it "knew" to monitor traffic information from the local Department of Transportation (DOT) the morning C. R. needed to get to the airport
    - The accident with a chemical spill that closed a road was also noted by thousands of other VPAs
    - The DOT service had the capability to negotiate with all those VPAs to come up with a plan to route traffic around the accident
    - All that C.R. had to do was realize why his wake-up alarm was earlier than expected, and
    - to follow his VPA's suggested detour to the airport that the DOT provided to the VPA

Examples of web services (CONTD…)

- for planning and taking a business trip (CONTD…)
  - C.R.'s VPA also interacted with his eyeglasses that were augmented with a heads-up display, an earpiece, and a camera
  - His VPA was able to "appear" whenever needed by accessing a myriad of services in the cloud so that it could help C. R.
  - negotiate the city streets,
  - avoid food allergies,
  - translate a menu from French to English,
  - learn more about the art he was viewing, and so on

Integrating services via Web Services

- the term service
  - From a technical perspective, the term service has been used to refer to any software function
  - that carries out a business task,
  - provides access to files (e.g., text, documents, images, video, audio, etc.), or
  - performs generic functions like authentication or logging

Integrating services via Web Services (CONTD…)

- Here we primarily focus on how services can be used
  - to share logical functions across different applications and
  - to enable software that runs on disparate computing platforms to collaborate

- Web services provide the means to
  - integrate disparate systems and
  - expose reusable business functions over HTTP

  - They either leverage HTTP
    - as a simple transport over which data is carried (e.g., SOAP/WSDL services) or
    - use it as a complete application protocol that defines the semantics for service behavior (e.g., RESTful services)

Terminology

- Web service developers often use different terms to refer to equivalent roles
- The first column lists names used to denote software processes that send requests or trigger events
- The second column contains terms for software functions that respond or react to these requests and events
- The terms appearing under each column are therefore synonymous

• Client	Service
• Requestor	Provider
• Service consumer	Service provider

Why Use Web Services?

- Web services make it relatively easy
  - to reuse and
  - share common logic with such diverse clients as mobile, desktop, and web applications

- The broad reach of web services is possible because they rely on open standards that are
  - ubiquitous,
  - interoperable across different computing platforms, and
  - independent of the underlying execution technologies
- All web services, at the very least, use HTTP and leverage data-interchange standards like XML and JSON, and common media types
- web services use HTTP in two distinct ways
  - Some use it as an application protocol to define standard service behaviors
  - Others simply use HTTP as a transport mechanism to convey data

Why Use Web Services? (CONTD…)

- Web services facilitate rapid application integration
  - because, when compared to their predecessors, they tend to be much easier to learn and implement
- web services facilitate the creation of complex business processes through service composition,
  - Due to their inherent interoperability and simplicity,
  - This is a practice in which compound services can be created by assembling simpler services into workflows
- Web services establish layers that naturally insulates clients from the means used to fulfill their requests
  - This makes it possible for clients and services to evolve somewhat independently as long as breaking changes do not occur on the service's public interface

Web Service Considerations

- problem of latency
  - Web services are “expensive” to call
  - Clients must serialize all input data to each web service (i.e., the request) as a stream of bytes and transmit this stream across computer processes
  - The web service must deserialize this stream into a data format and structure it understands before executing
  - If the service provides a “complex type” as a response (i.e., something more than a simple HTTP status code), then
  - the web service must serialize and transmit its response, and
  - the client must deserialize the stream into a format and structure it understands
  - All of these activities take time (problem of latency)

Web Service Considerations (CONTD…)

- partial failures
  - Possibly more important than the problem of latency is the fact that web service calls typically entail distributed communications
  - This means that client and service developers alike must be prepared to handle partial failures
  - A partial failure occurs when the client, service, or network itself fails while the others continue to function properly
  - Networks are inherently unreliable, and problems may arise for innumerable reasons
    - Connections will occasionally time out or be dropped
    - Servers will be overloaded from time to time, and as a result, they may not be able to receive or process all requests
  - Services may even crash while processing a request
  - Clients may crash too, in which case the service may have no way to return a response
  - Multiple strategies must therefore be used to detect and handle partial failures

Web Service Considerations (CONTD…)

- Web services should therefore be reserved for situations when:
  - The client and service belong to different application domains and the "service functions" cannot be easily imported into the client
  - The client is a complex business process that incorporates functions from multiple application domains
  - The logical services are owned and managed by different organizations and charge at different rates
  - The divide between the client and server is natural
    - The client may, for example, be a mobile or desktop application that uses common business functions

Web Service Considerations (CONTD…)

- Services and the Promise of Loose Coupling
  - Services are often described as being loosely coupled
  - Coupling is the degree to which some entity (e.g., client) depends on another entity
    - When the dependencies are many, the coupling is said to be high or tight (e.g., high coupling, tightly coupled)
    - Conversely, when the dependencies are few, coupling is considered to be low or loose (e.g., low coupling, loosely coupled)
  - It is certainly true that web services can eliminate the client's dependencies on the underlying technologies used by a service

Web Service Considerations (CONTD…)

- Services and the Promise of Loose Coupling (CONTD…)
  - The following list describes a few forms of coupling that service designers must consider
    - Function coupling:
      - Clients expect services to consistently produce certain results given certain types of input under particular scenarios
      - Clients are therefore indirectly dependent on the logic implemented by web services
      - The client will most certainly be affected if this logic is implemented incorrectly or
      - is changed to produce results that are not in accordance with the client's expectations

Web Service Considerations (CONTD…)

- Services and the Promise of Loose Coupling (CONTD…)
  - Data structure coupling:
  - Clients must understand the data structures that a service receives and returns, the data types used in these structures, and the character encodings (e.g., Unicode) used in messages
  - If a data structure provides links to related services, the client must know how to parse the structure for that information
  - The client may also need to know what HTTP status codes the service returns
  - Service developers must be careful to refrain from including platform-specific data types (e.g., dates) in data structures

Web Service Considerations (CONTD…)

- Services and the Promise of Loose Coupling (CONTD…)
  - Temporal coupling:
  - A high degree of temporal coupling exists when a request must be processed as soon as it's received
  - The implication is that the systems (e.g., databases, legacy or packaged applications, etc.) behind the service must always be operational
  - Temporal coupling can be reduced if the time at which a request is processed can be delayed
  - Web services can achieve this outcome with the Request/Acknowledge pattern
  - Temporal coupling is also high if the client must block and wait for a response
  - Clients may use the Asynchronous Response Handler pattern to reduce this form of coupling

Web Service Considerations (CONTD…)

- Services and the Promise of Loose Coupling (CONTD…)
- URI coupling:
- Clients are often tightly coupled to service URIs
- That is, they often either have a static URI for a service, or follow a simple set of rules to construct a service URI
- Unfortunately, this can make it difficult for service owners to move or rename service URIs, or to adopt new patterns for URI construction since actions like these would likely cause clients to break

Service-Oriented Architecture Overview

- The business trip that C. R. took involved using multiple services, both inside and outside his organization, such as travel, car rental, online calendar, and customer relationship management (CRM) services
- From a software architectural point-of-view, this is a service-oriented architecture (SOA)
- An SOA is built using a collection of services that communicate with each other
- The communication can involve either simple data passing or it could involve two or more services coordinating some activity
- Some means of connecting services to each other is needed
- Those connections are Web services

Service-Oriented Architecture Overview (CONTD…)

- The inception of the Web service technology is a specific implementation of SOA
- It uses
  - XML as a common message format
  - Other standard Internet protocols, such as Hypertext Transfer Protocol (HTTP) and HTTP Secure (HTTPS),
- This has sparked new interest in this type of architecture, because
  - using these standards is simple,
  - is universally supported, and
  - does not require anyone to reinvent the wheel

History of Web Services Specification

- Originally the only Web services specification included
  - the Web Services Description Language (WSDL);
  - Universal Description, Discovery, and Integration (UDDI): and
  - SOAP
- Later
  - SOAP/WSDL without UDDI,
  - REST,
  - XML, and
  - JSON

History of Web Services Specification

- Web Services Description Language
  - WSDL forms the basis for the original Web services specification
  - Figure 3.3 below illustrates the use of WSDL
  - At the left is a service provider and at the right is a service consumer
  - The steps involved in providing and consuming a service are as follows:
    1. A service provider describes its service using WSDL. This definition is published to a registry of services. The registry uses UDDI.
    2. A service consumer issues one or more queries to the registry to locate a service and determine how to communicate with that service.
    3. Part of the WSDL provided by the service provider is passed to the service consumer. This tells the service consumer what the requests and responses are for the service provider.
    4. The service consumer uses the WSDL to send a request to the service provider.
    5. The service provider provides the expected response to the service consumer.

History of Web Services Specification (CONTD…)

- Web Services Description Language (CONTD…)

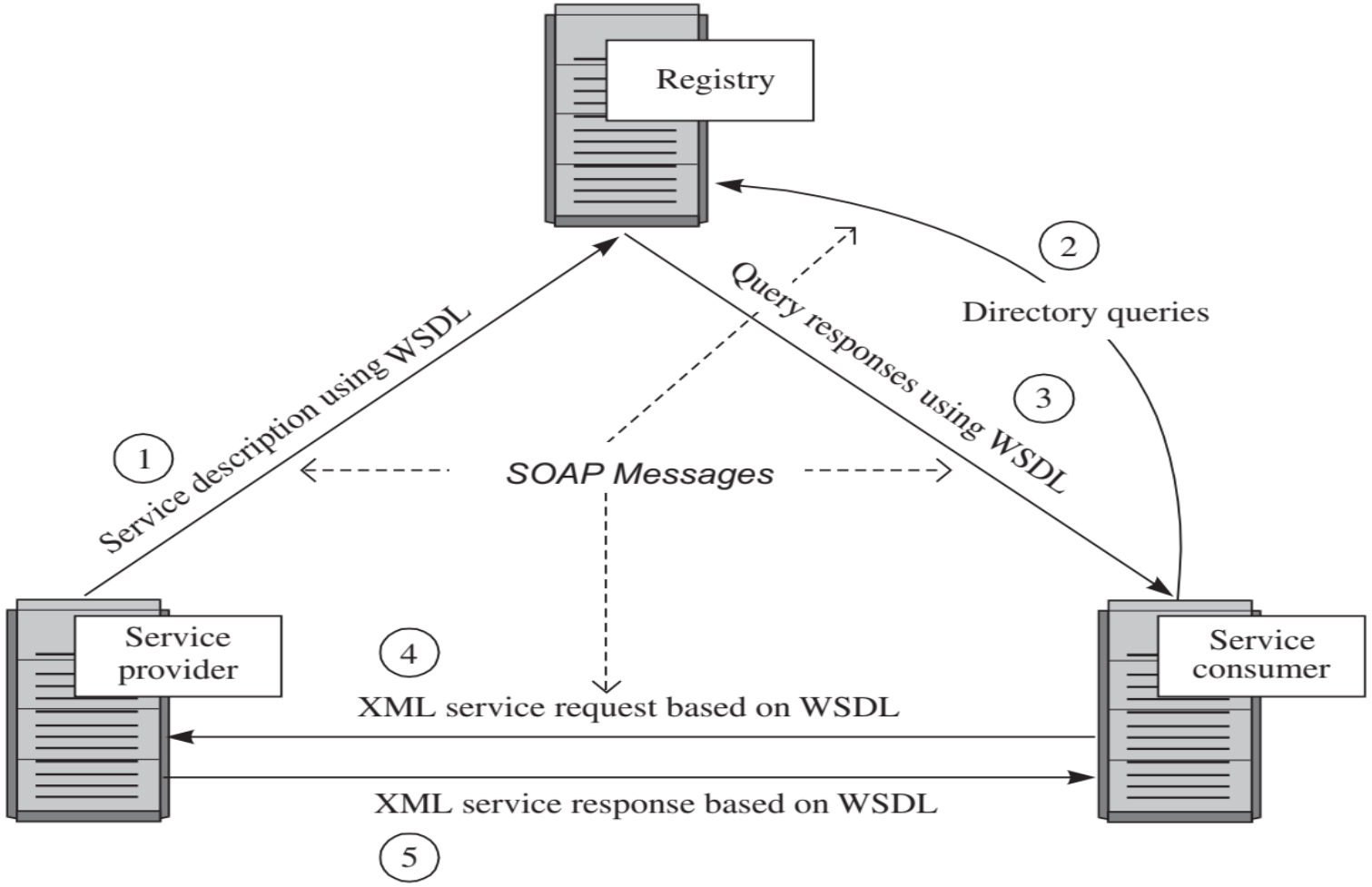


Figure 3.3 Web services basics.

History of Web Services Specification (CONTD…)

- Universal Description, Discovery, and Integration
  - The UDDI registry was intended to serve as a means of "discovering" Web services described using WSDL
  - The idea was that the UDDI registry could be searched in various ways to obtain contact information and the services available from various organizations
  - UDDI registries have not been widely implemented
  - The term registry is sometimes used interchangeably with the term service repository
  - Generally, repositories contain more information than a strict implementation of a UDDI registry
  - Today, instead of active discovery, repositories are used mainly at design time and to assist with governance

History of Web Services Specification (CONTD…)

- SOAP
- All the messages shown in Figure 3.3 above are sent using SOAP
- (SOAP at one time stood for Simple Object Access Protocol; now the letters in the acronym have no particular meaning)
- SOAP provides the envelope for sending Web services messages
- SOAP generally uses HTTP
  - HTTP is the familiar connection we all use for the Internet
- Figure 3.4 below provides more detail on the messages sent using Web services
- At the left of the figure is a fragment of the WSDL sent to the registry
- It shows a CustomerInfoRequest that requires the customer's account to object information

History of Web Services Specification (CONTD…)

- SOAP (CONTD…)

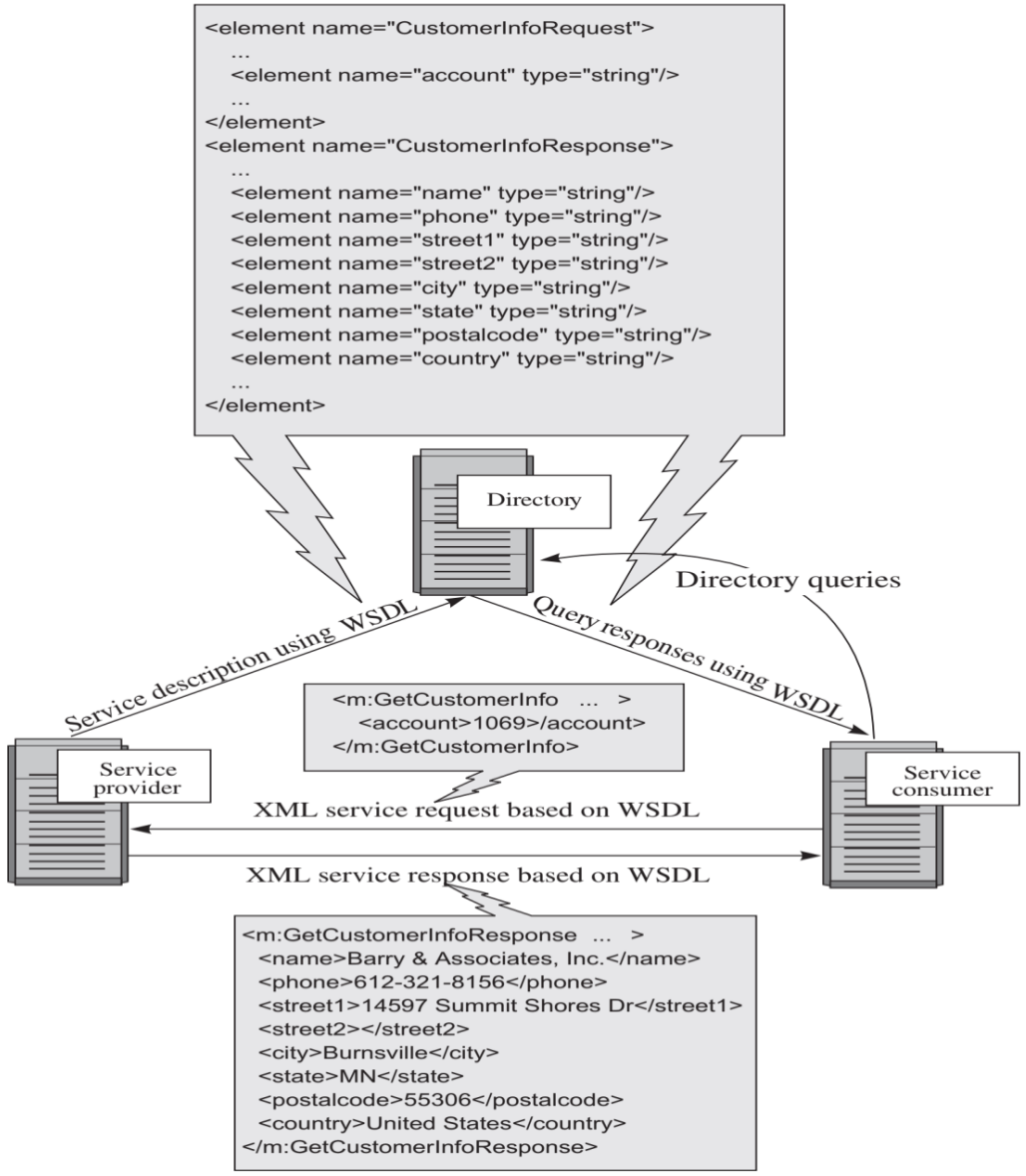


Figure 3.4 SOAP messaging with a directory.

History of Web Services Specification (CONTD…)

- Using SOAP without UDDI
  - The interaction between the service provider and the service consumer is shown in Figure 3.5
  - This is the nature of virtually all SOAP Web services today

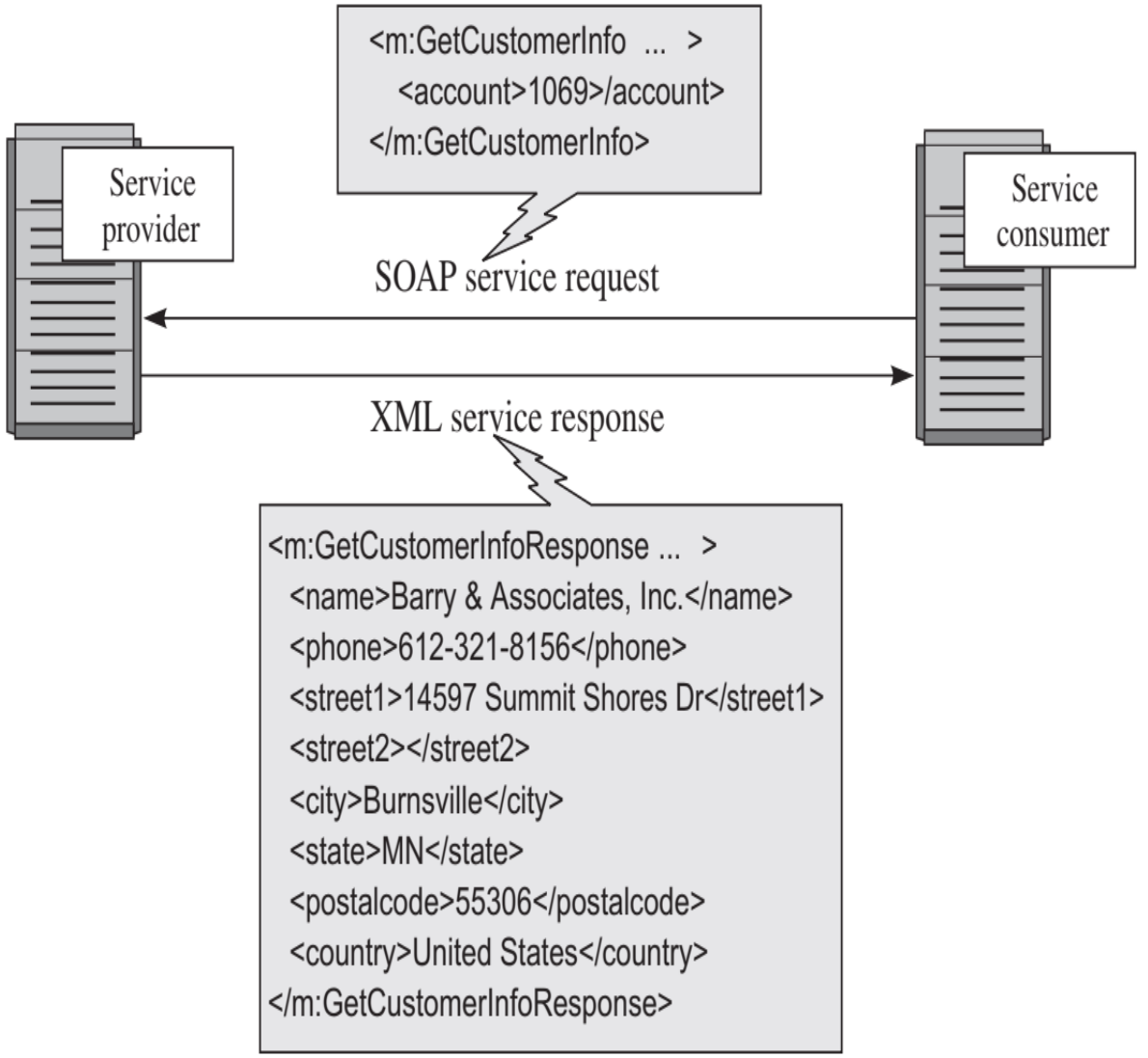


Figure 3.5 SOAP messaging.

History of Web Services Specification (CONTD…)

- Using REST
- The first alternative to SOAP that was developed is Representational State Transfer (REST)
- REST is a style of architecture based on a set of principles that describe how networked resources are defined and addressed
- Roy Fielding first described these principles in 2000 as part of his doctoral dissertation
- REST appeals to developers because
  - it has a simpler style that makes it easier to use than SOAP,
  - is a bit less verbose than SOAP (sends less down the “wire”), and
  - is used in a way that other resources are used on the Internet
- Figure 3.6 below illustrates a fragment of a REST message
- It looks a lot like any other HTTP request that uses parameters
- The return message in this example looks much like the return messages from SOAP

History of Web Services Specification (CONTD…)

- Using REST (CONTD…)

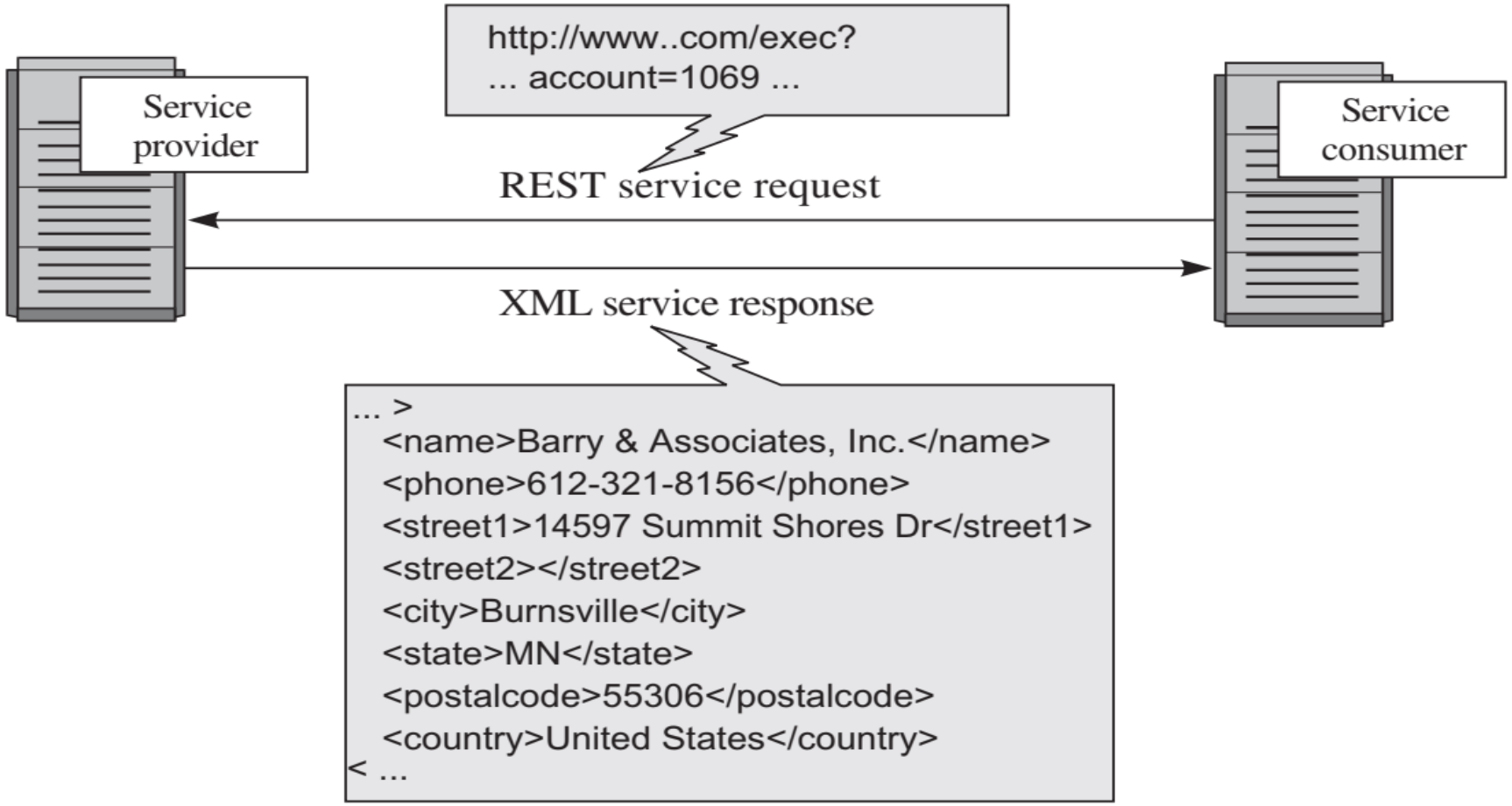


Figure 3.6 REST messaging.

History of Web Services Specification (CONTD…)

- Using XML
- The examples here show both SOAP and REST using XML for response messages
- XML has a tagged message format
- This is shown in Figure 3.7
- The tag <city> is highlighted in this figure
- The value of city is Burnsville
- The tag </city> is the ending tag indicating the end of the value of city
- Both the service provider and service consumer use these tags
- In fact, the service provider could send the data shown at the bottom of Figure 3.7 in any order
- The service consumer uses the tags and not the order of the data to get the data values

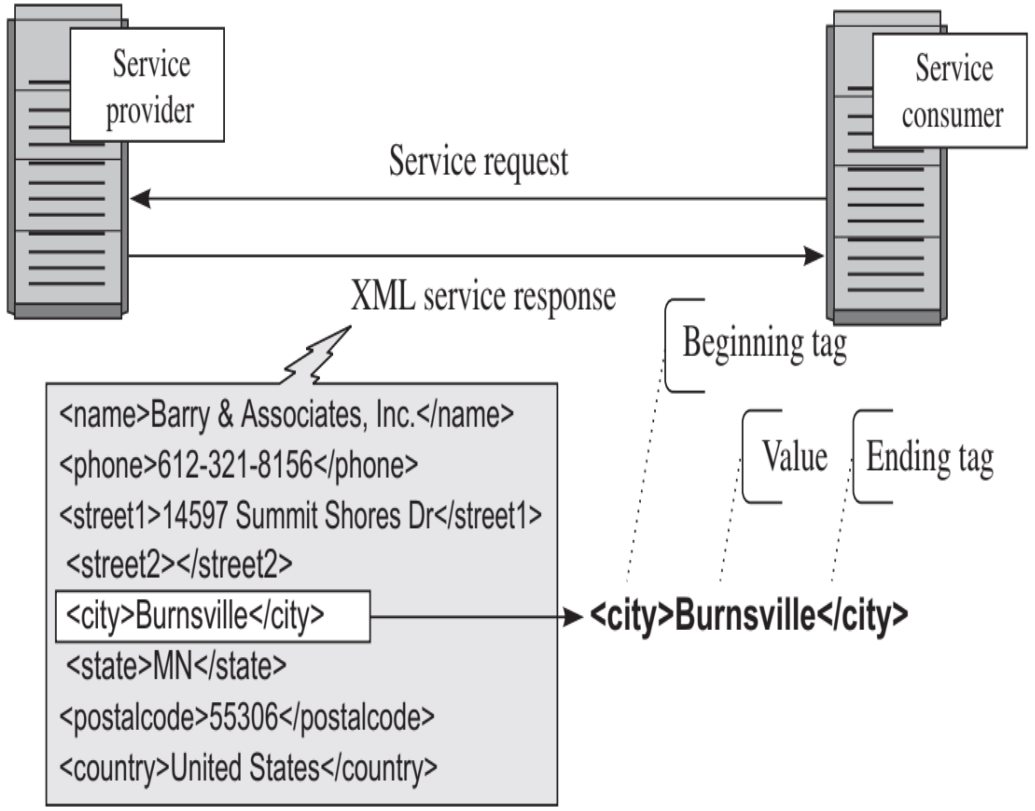


Figure 3.7 Tagged messages.

History of Web Services Specification (CONTD…)

- Using XML (CONTD…)
- The XML-tagged format provides a level of resilience not available with fixed record formats commonly used before the advent of XML
- For example, if a service provider adds an additional element not expected by a service consumer, the XMLtagged format allows processing to continue without any problems occurring
- What if the data sent changes when using XML?
- Figure 3.9 shows that a service provider has added a new element, <extension> for a telephone extension

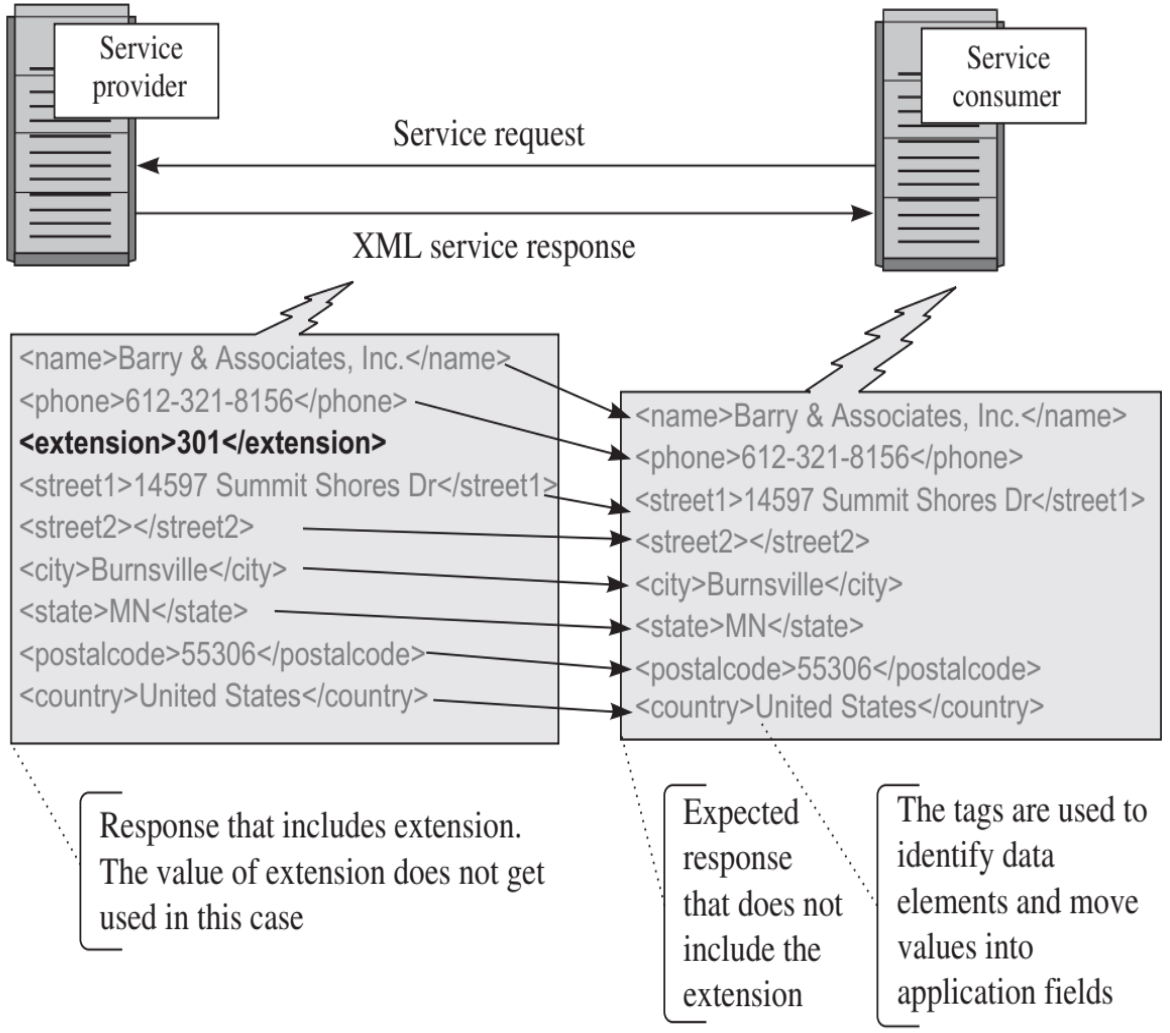


Figure 3.9 Example of the resilience provided by tagged messages.

History of Web Services Specification (CONTD…)

- JSON, an XML Alternative
- It is possible to use Web services without XML
- JSON (JavaScript Object Notation) is one option
- It uses name/value pairs instead of the tags used by XML
- For example, the name “city” is paired with the value “Burnsville”
- This is illustrated on the right side of Figure 3.13
- The name/value pairs in JSON provide the same type of resilience as the XML-tagged format for data exchanges described in the previous section
- The name/value pairs do not have to be in any particular order to work

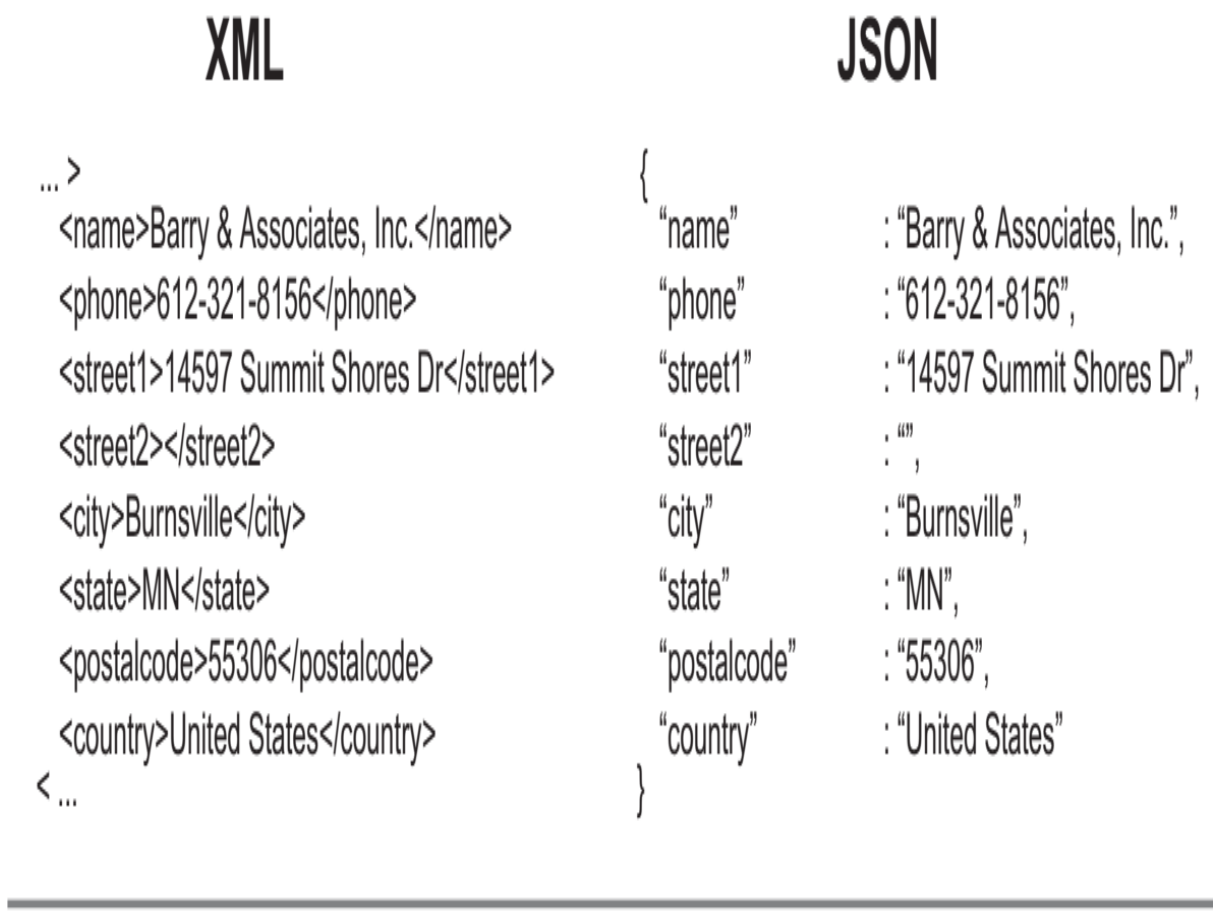


Figure 3.13 Comparison of XML and JSON.

When to Use SOAP, REST, JSON, or Other Options

- If you are developing your own service, you can choose the Web service that is best for you

- The one that is best for you might be
  - the Web service used by most in your industry or
  - the Web service used by most services on the Internet that you are most likely to use

- Be prepared, however, to use "all of the above" as mentioned before

- However, there may be technical reasons that you should choose one Web service over another

RELATIONSHIP OF WEB SERVICES AND SOA

- Figure 3.14 uses a Venn diagram to illustrate the relationship between SOA and Web services
- The overlapping area in the center represents SOA using Web services for connections
- The nonoverlapping area of Web services represents that Web services can be used for connections, but connections alone do not make for an SOA
- The non-overlapping area of SOA indicates that an SOA can use Web services as well as connections other than Web services (the original specifications of CORBA and DCOM are examples)

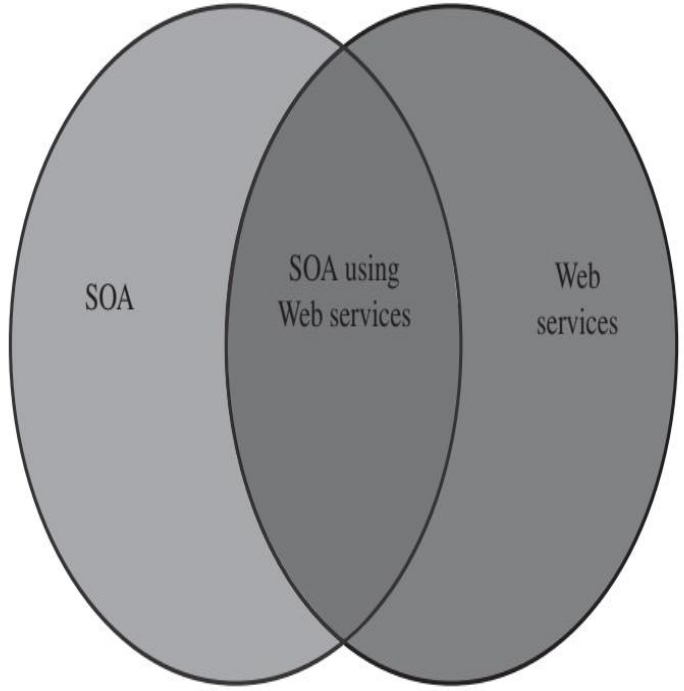


Figure 3.14 Relationship of Web services and SOA.

IDENTIFICATION AND DESIGN OF SERVICES

- Key to SOA is the identification and design of services
- The idea is that services should be designed in such a way that they become components that can be assembled in multiple ways to support or automate business functions
- It is not necessarily easy to properly identify and design services
- When done well, the services allow an organization to quickly assemble services—or modify the assembly of services—to add or modify the support or automation of business functions

IDENTIFICATION AND DESIGN OF SERVICES (CONTD)

- Services are assembled to support or automate business functions
- Figure 3.16 below illustrates the assembly of services
- This represents an SOA
- Web services are used to connect the services in an SOA
- The services in an SOA can come from any service provider (which, as mentioned earlier, can also be a service consumer)
- So, in a given SOA, the services might be from internal systems along with any number of external systems accessible anywhere on the Internet
- This is illustrated by Figure 3.17 below
- It is easy to imagine that you can reassemble the same services with other services to achieve a different functionality
- This ability to change the assembly of services is one way that an SOA can quickly adapt to changing business needs

IDENTIFICATION AND DESIGN OF SERVICES (CONTD)

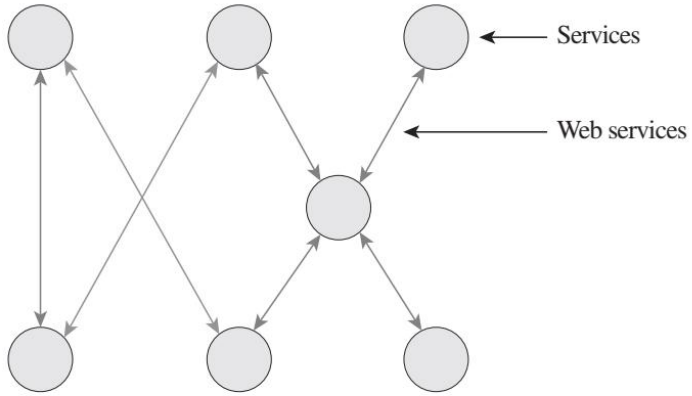


Figure 3.16 Assembly of services into an SOA.

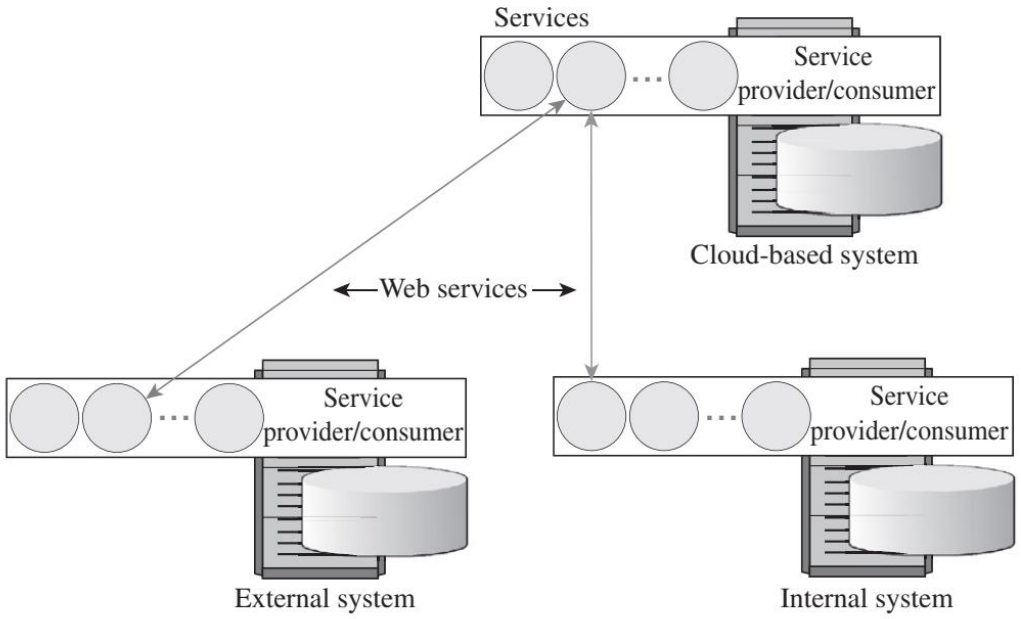


Figure 3.17 Example sources of services in an SOA.